

# Snare Parameter Guide

Parameter reference — Thee Coast Machine

## 1. Internal Structure

A three-layer hybrid: **noise + tonal oscillator + snap burst**. Follows the TR-808 / 909 model, with the Crisp parameter taking you from full lo-fi to a sharp modern transient.

| Element                  | Role                                                         | Sonic character  |
|--------------------------|--------------------------------------------------------------|------------------|
| Noise (White / Pink)     | Body source. NOISE selects the colour.                       | "shh" / "shhk"   |
| Tonal oscillator         | Fast pitch sweep 500 → 200 Hz over ~40 ms — provides body    | "thock" / "doom" |
| Snap burst               | Very short white-noise pulse right at the transient (1–5 ms) | "kah" / "pah"    |
| 1-pole HP filter (Crisp) | Highpass on the noise layer (30 Hz – 5 kHz)                  | lo-fi vs. crispy |

## 2. Parameter Reference

### DECAY (0.05 — 1.5 sec)

Total **sustain time**.

- **0.05–0.10**: tight, tick-like
- **0.15–0.20**: standard snare
- **0.25–0.40**: full, sustained
- **0.50+**: barely a snare anymore

**Tip**: Driven by tempo and space.

### NOISE (0 = White, 1 = Pink)

**Noise colour**.

- **White (0)**: flat / bright → 909 / Modern
- **Pink (1)**: tilted low / warm → 808 / Vintage

**Tip**: Modern: white. Warm: pink.

### TONE (0 — 1)

**Mix amount** of the tonal sine oscillator.

- **0.0**: pure noise (clap-like)
- **0.3–0.5**: balanced body
- **0.7+**: tom-like

- **1.0**: basically a tom

**Tip:** 0.4–0.5 is standard.

### **SNAP** (0 — 1)

**Very short noise burst** at the trigger.

- **0.0**: smooth attack
- **0.3–0.5**: natural emphasis
- **0.8–1.0**: sharp, in your face

**Tip:** Raise to cut through a busy mix.

### **CRISP** (0 = Lo-Fi, 1 = Crisp)

**1-pole highpass filter** on noise.

- **0.0**: 30 Hz cut (bypass)
- **0.3**: ~450 Hz
- **0.5**: ~1250 Hz
- **0.7**: ~2500 Hz
- **1.0**: 5 kHz cut

**Tip:** Lo-Fi: stay low. Modern: 0.5+.

### 3. Character Presets

| Goal                    | DEC  | NOISE | TONE | SNAP | CRISP |
|-------------------------|------|-------|------|------|-------|
| TR-808 (warm)           | 0.15 | 0.65  | 0.40 | 0.45 | 0.20  |
| TR-909 (House / Techno) | 0.18 | 0.35  | 0.35 | 0.65 | 0.45  |
| TR-606 (tight)          | 0.08 | 0.50  | 0.50 | 0.55 | 0.60  |
| Acid (sharp)            | 0.12 | 0.40  | 0.45 | 0.70 | 0.55  |
| Industrial              | 0.15 | 0.30  | 0.55 | 0.75 | 0.65  |
| Dub (soft, long)        | 0.30 | 0.70  | 0.40 | 0.20 | 0.05  |
| Lo-Fi / Boom Bap        | 0.10 | 0.40  | 0.25 | 0.35 | 0.00  |
| DnB / Jungle            | 0.10 | 0.40  | 0.45 | 0.75 | 0.65  |
| Modern Tech House       | 0.13 | 0.45  | 0.45 | 0.60 | 0.55  |
| Ambient                 | 0.25 | 0.70  | 0.30 | 0.15 | 0.05  |

### 4. Tuning Order

| #      | Do this                      | Why                      |
|--------|------------------------------|--------------------------|
| Step 1 | Set DECAY                    | How long should it ring? |
| Step 2 | Balance TONE                 | Want body, or not?       |
| Step 3 | Adjust brightness with NOISE | White vs. Pink           |
| Step 4 | Cut-through with CRISP       | Muffled, or up front?    |
| Step 5 | Finish with SNAP             | Fine-tune attack feel    |

### 5. Troubleshooting

| Symptom                   | Cause & fix                                            |
|---------------------------|--------------------------------------------------------|
| Thin / weak snare         | Raise TONE (0.4–0.6) and SNAP.                         |
| Clashes with the kick     | Raise CRISP to highpass the noise away from the kick.  |
| Too much "shh"            | Move NOISE toward pink, lower CRISP, or shorten DECAY. |
| Sounds like a tom         | Lower TONE below 0.5.                                  |
| Weak attack               | Push SNAP to 0.5+, raise CRISP.                        |
| Stands out on slow tracks | Pull SNAP and CRISP back, extend DECAY.                |
| Distortion on rolls       | Lower SnareVol or shorten DECAY.                       |

※ The snare carries the backbeat. Together with the kick it defines the groove. When layered with the Clap, separate by PAN or FREQ.